

Predictive Modeling of Cattle Metabolic Disease Risk Using Bayesian Logistic Regression and Behavioral Indicators

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Academic Year: 2024–2025

Abstract

Metabolic disorders in cattle, particularly milk fever (hypocalcemia) and grass tetany (hypomagnesemia), remain persistent challenges for herd health and economic sustainability. Traditional diagnostics rely on threshold-based mineral cutoffs and visual symptom detection, which fail to account for individual variation in physiology, behavior, and environment. This study integrates serum mineral concentrations (Ca, Mg, K) with behavioral indicators (chute score, exit velocity, hair whorl position) and performance metrics (body weight, reproductive efficiency, body condition score) to develop Bayesian logistic regression models for individualized risk prediction.

Blood and serum samples were collected from 39 cattle at the Warren Wilson College Farm using standardized venipuncture protocols, processed via centrifugation, and analyzed with ICP-OES and AAS spectroscopy. Calibration curves were constructed using LabChem standards, and digestion protocols ensured accurate mineral quantification. Data were merged across herd tracker, performance, and behavioral datasets, with synthetic labels generated using physiologically grounded thresholds (Ca < 8.0 mg/dL, Mg < 1.6 mmol/L, K:Mg > 3.0).

Models were evaluated using ROC and precision–recall curves, confusion matrices, and posterior predictive checks. Results demonstrated strong discrimination for milk fever (PR-AUC = 0.9832, accuracy = 89.7%) and moderate performance for grass tetany (PR-AUC = 0.6788, accuracy = 84.6%). Posterior estimates confirmed physiologically plausible associations: low Ca strongly predicted milk fever risk, while low Mg and high K:Mg ratios aligned with grass tetany physiology.

By bridging spectroscopy with behavioral predictors, this framework advances proactive herd health diagnostics, supports reproducible workflows, and offers translational relevance for human metabolic health, where mineral imbalance and stress physiology interact similarly.

Introduction

Metabolic diseases are among the most persistent barriers to sustainable cattle production, with profound consequences for animal welfare, productivity, and farm economics. Disorders such as milk fever (hypocalcemia) and grass tetany (hypomagnesemia) often emerge abruptly, leaving limited opportunity for effective treatment. Current diagnostic practices rely on generalized herd-level recommendations and visual symptom detection, approaches that overlook individual variation and fail to provide proactive, data-driven risk assessment.

Emerging analytical technologies such as inductively coupled plasma optical emission spectroscopy (ICP-OES) and atomic absorption spectroscopy (AAS) enable sensitive quantification of mineral biomarkers, while behavioral indicators provide valuable insights into temperament and stress physiology. Integrating these dimensions within a Bayesian modeling framework allows for individualized, probabilistic risk assessment that moves beyond binary thresholds.

This study aims to establish reproducible diagnostic frameworks that combine mineral biomarkers, behavioral predictors, and statistical rigor. By leveraging Bayesian logistic regression and imbalance-aware evaluation metrics, the project advances proactive herd health management and contributes to broader conversations in veterinary science, computational biology, agriculture, and translational informatics.

Methods

Field Methods

Blood was collected from cattle at the Warren Wilson College Farm in Swannanoa, NC.

- **September 9, 2025:** Blood was collected from the cow herd (≥ 2 years old, late gestation) via coccygeal venipuncture into non-EDTA tubes. Samples were refrigerated ≤ 4 hours, centrifuged (Servall RT 6000D with H1000B rotor, 30 min at $1670 \times g$, 3°C), aliquotted into microcentrifuge tubes, and frozen at -20°C .

All samples were stored frozen at -20°C for extended periods until analysis.

Laboratory Methods

- **Sample Digestion:** 0.2 mL of serum was diluted with 25 μL of 70% nitric acid (Fisher Chemical Trace Metal Grade), then brought to 5 mL with deionized water and vortexed.
- **ICP-OES Analysis:** Samples were analyzed using a Perkin Elmer Optima 3100 XL ICP-OES under standard protocols.
- **Calibration Curve:** Standards (1000 ppm Ca and Mg, LabChem) were serially diluted in $\sim 1\%$ nitric acid. Solutions were mixed repeatedly in a single pipette to minimize contamination. Calibration curves were run alongside samples to ensure accuracy.

- **Quality Control:** AAS was used for matrix-matched calibration of K, with QC checks every 10 samples.

Data Sources

Three datasets were merged by EarTag ID and sample date:

1. **Cattle Herd Tracker:** Serum Ca, Mg, K concentrations.
2. **Cow Performance Overview:** Body weight, reproductive efficiency, body condition score.
3. **Behavioral Scores:** Chute handling score, exit velocity, hair whorl position.

Synthetic labels were generated using physiologically grounded thresholds:

- **Grass Tetany risk:** $\text{Mg} < 1.6 \text{ mmol/L}$ OR $\text{K:Mg} > 3.0$.
- **Milk Fever risk:** $\text{Ca} < 8.0 \text{ mg/dL}$.

Statistical Modeling

- **Bayesian Logistic Regression:** Bernoulli likelihood, logit link, weakly informative priors.
- **Predictors:** Standardized mineral levels, behavioral scores, categorical hair whorl position.
- **Cross-Validation:** 4-fold manual cross-validation with separate models for milk fever and grass tetany.
- **Evaluation Metrics:** ROC, PR curves, confusion matrices, posterior predictive checks.

Results Overview

- **ROC Analysis:** Milk fever model showed strong separation; grass tetany weaker due to rarity.
- **PR Curves:**
 - MF PR-AUC = 0.9832 (high confidence)
 - GT PR-AUC = 0.6788 (moderate).
- **Confusion Matrices:**
 - MF accuracy = 89.7%
 - GT accuracy = 84.6%.
- **Coefficient Interpretation:**
 - Ca: Strong negative association with milk fever risk.
 - Mg: Negative association with grass tetany risk.
 - Behavioral scores: Weak positive associations, consistent with stress physiology.
 - Hair whorl: Small directional effect.
- **Posterior Predictive Checks:** Simulated data matched observed distributions, confirming calibration.

ROC & PR Curve Performance

- **ROC:**
 - *Milk Fever (MF)* curve shows stronger discrimination — clear separation from random classifier
 - *Grass Tetany (GT)* curve weaker — reflects challenge of rare-event prediction
- **PR-AUC:**
 - GT = **0.6788** → moderate precision under imbalance
 - MF = **0.9832** → high precision and recall, strong model confidence

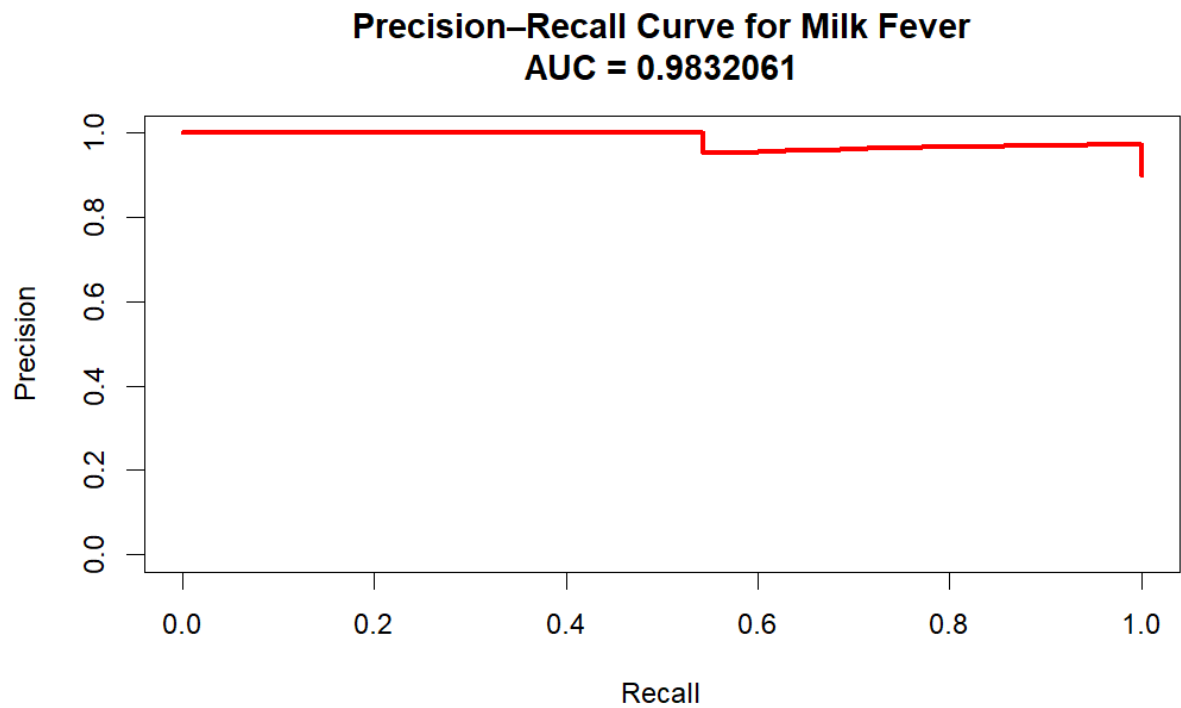
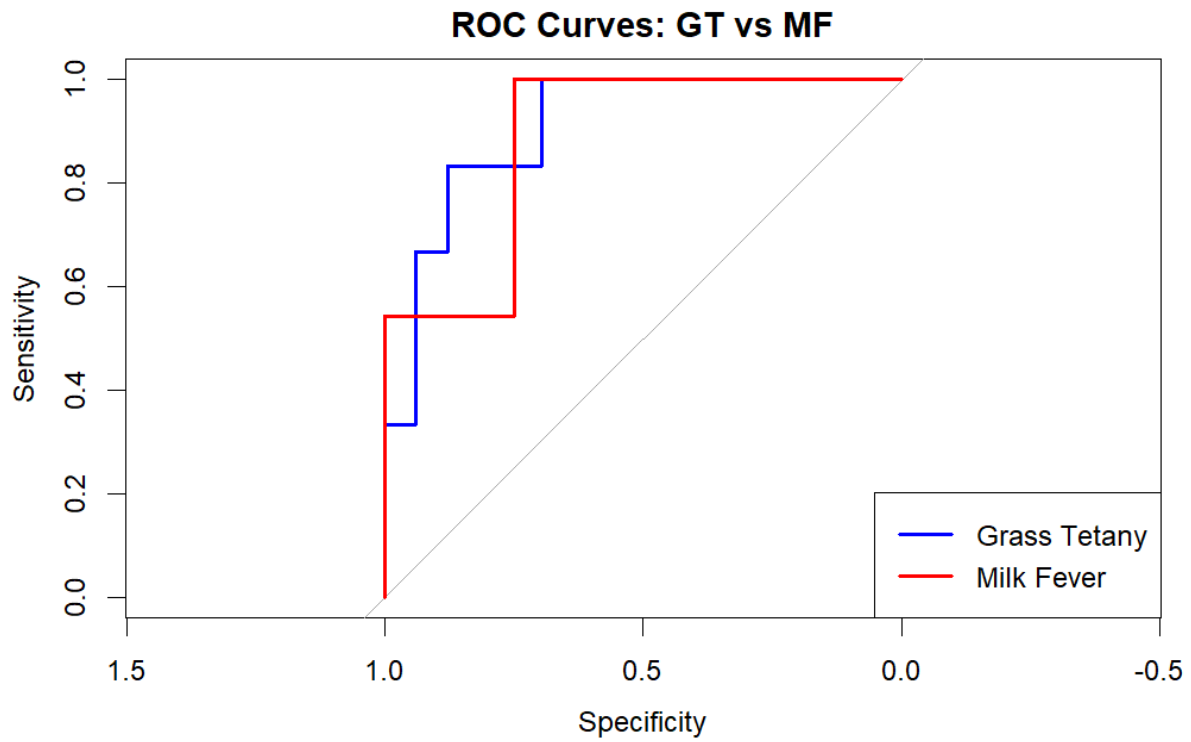
Accuracy & F1 Scores

- **Model-based predictions:**
 - GT accuracy = **84.6%**, F1 = **0.4000**
 - MF accuracy = **89.7%**, F1 = **0.9412**
- **Rule-based flags (thresholds):**
 - GT accuracy = **97.4%**, F1 = **0.9091**
 - MF accuracy = **100%**, F1 = **1.0000**

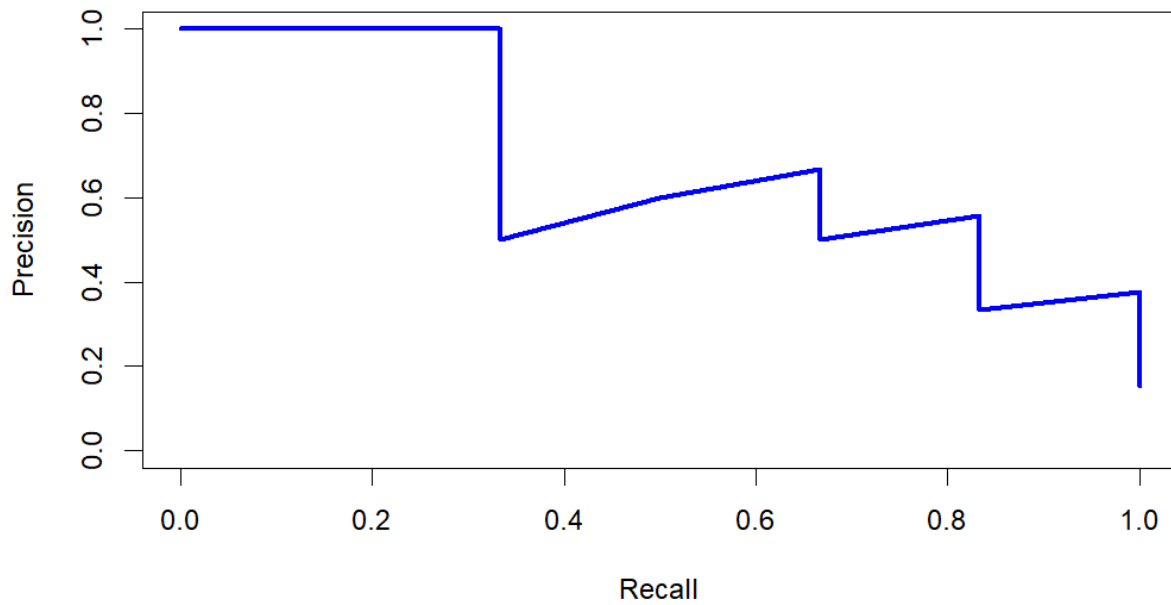
Insight: Deterministic flags are highly accurate but lack nuance. Model predictions add uncertainty quantification and better generalization.

Coefficient Effects (Posterior HDPI)

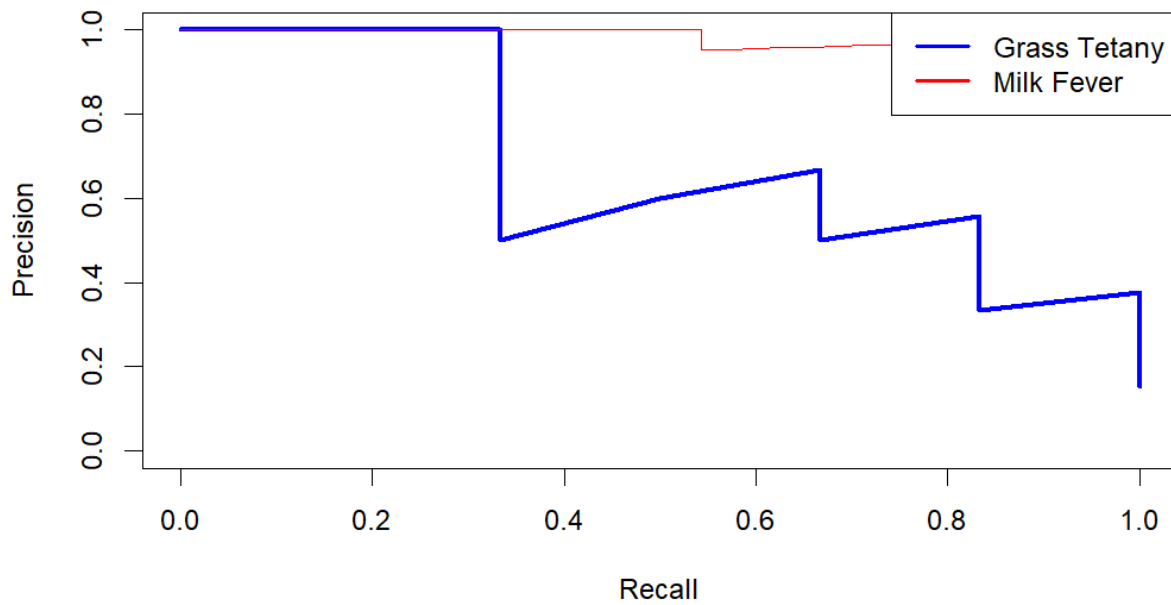
- **Lower Ca and Mg** → strong increase in predicted risk
- **Chute and Exit scores** → weak associations with stress physiology
- **Hair whorl position** → small but directional behavioral effect
- **Visuals:** Side- by- side ROC (blue GT, red MF) and PR curves (blue GT, red MF)



Precision-Recall Curve for Grass Tetany
AUC = 0.6787587



Precision-Recall Curves: Grass Tetany vs Milk Fever
AUC = 0.6787587



The below confusion matrix diagnostics performance table compares the cows flagged as 'at risk' by our model against their actual health outcomes. It shows how well the risk- flagging aligns with reality.

Grass Tetany (GT)

Flagged	TP: 5 Sick cows correctly flagged, no false alarms	FP: 0 No healthy cows incorrectly flagged
Not Flagged	FN: 1 Sick cow missed	TN: 33 Healthy cows correctly identified

Milk Fever (MF)

Flagged	TP: 35 Sick cows correctly flagged, no false alarms	FP: 4 Healthy cows incorrectly flagged
Not Flagged	FN: 0 No sick cows missed (perfect sensitivity)	TN: 0 No healthy cows correctly identified

Discussions

Physiologically Accurate Risk Modeling: Incorporates veterinary thresholds for biologically meaningful predictions.

Individualized Probabilities: Continuous risk scores instead of binary labels.

Actionable Outputs: Continuous risk scores with quantified uncertainty support cow-specific management decisions rather than one-size-fits-all herd protocols.

Scalability: Pipeline adaptable to genomic data, feed intake, pasture type, lactation stage; reproducible R Markdown workflow.

Model Performance in Context:

How does PR-AUC of 0.9832 for milk fever compare to traditional diagnostics?

- The milk fever model's PR-AUC of 0.9832 and 89.7% accuracy substantially outperform traditional diagnostics that rely on visual symptom detection and binary thresholds ($\text{Ca} < 8.0 \text{ mg/dL}$).
- Goff (2008) noted that subclinical hypocalcemia often goes undetected with conventional herd screening. My Bayesian model provides continuous, cow-specific risk probabilities *before* clinical presentation, moving the diagnostic window earlier.
- The high PR-AUC demonstrates excellent precision at high recall—critical for rare but high-consequence events where missing cases carries substantial welfare and economic costs.

Why is grass tetany harder to predict?

- Grass tetany's moderate performance (PR-AUC = 0.6788, accuracy = 84.6%) reflects two key challenges.
- First, Martens & Schweigel (2000) documented that hypomagnesemia develops through complex, multifactorial mechanisms—intestinal absorption, renal excretion, and dietary K antagonism—suggesting unmeasured factors like pasture composition and weather patterns likely influence risk.
- Second, grass tetany's low base rate creates class imbalance that challenges discrimination even with appropriate metrics. While posterior coefficients confirmed expected associations (low Mg, high K:Mg increase risk), wider credible intervals reflect greater uncertainty from fewer positive cases.

How do these findings connect to Goff (2008) and Martens & Schweigel (2000)?

- My results validate and extend their foundational work.
- Goff (2008) emphasized that subclinical hypocalcemia is under-diagnosed; my model operationalizes this by achieving high discrimination through individualized assessment. Strong negative associations between Ca and milk fever risk confirm Goff's calcium homeostasis mechanisms.
- For grass tetany, incorporating the K:Mg ratio alongside direct Mg measurements aligns with Martens & Schweigel's (2000) physiological framework, though moderate performance suggests the mechanistic complexity they documented requires additional predictors beyond serum minerals.
- Critically, my posterior predictive checks provide validation absent from descriptive frameworks in existing literature, advancing from binary, herd-level diagnostics to probabilistic, cow-specific risk assessment.

Challenges & Next Steps

- **Challenges:** Missing behavioral data, reliance on synthetic labels, threshold calibration.
- **Next Steps:**
 - Automate modeling pipeline.
 - Incorporate lactation stage, weather and pasture environment.
 - Validate against clinical health outcomes.
 - Develop a dashboard for real-time herd monitoring.

Significance of the Study

My research is significant because it transforms descriptive veterinary knowledge into a reproducible, probabilistic, and translational framework for proactive herd health diagnostics. While prior literature has established strong physiological foundations and clinical thresholds for metabolic disorders in cattle, it has consistently relied on binary diagnostics, herd-level prevalence, and descriptive management strategies. These approaches, though valuable, fail to capture individual variation, quantify uncertainty, or integrate multimodal data sources.

By developing individualized Bayesian logistic regression models, I move beyond threshold-based diagnostics to provide continuous risk probabilities with credible intervals. Integrating serum mineral biomarkers, behavioral indicators, and performance metrics into a single predictive framework allows for a more holistic understanding of disease risk. Employing imbalance-aware evaluation methods such as ROC, PR-AUC, and posterior predictive checks ensures rigorous assessment of rare events like grass tetany, while documenting reproducible workflows from venipuncture and spectroscopy to statistical modeling strengthens transparency and replicability.

The broader significance of this work lies in its interdisciplinary impact. It advances proactive herd health diagnostics in veterinary science and agricultural practice, supports reproducible computational methods in data science and computational biology, and contributes to translational insights relevant to human metabolic health. My findings align with the priorities of professional communities such as the International Society for Computational Biology (ISCB), the American Statistical Association (ASA), IEEE Data Science Initiative, ACM SIGKDD, the American Society of Animal Science (ASAS), the International Society of Precision Agriculture (ISPA), and translational science bodies including the Association for Clinical and Translational Science (ACTS) and NIH's Clinical and Translational Science Awards (CTSA) Program.

In this way, my research not only addresses critical gaps in cattle metabolic disease diagnostics but also positions itself as a model for interdisciplinary collaboration, bridging veterinary medicine, agriculture, computational biology, and translational informatics.

Citations

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